Enhancing Human Safety through Early Detection of Floods and Landslides in Sri Lanka.

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***Abstract*— In Sri Lanka, natural disasters particularly floods and landslides pose significant threats to public safety as well as the economic stability. This research introduces an innovative approach to improve disaster response by utilizing machine learning (ML) and Internet of Things (IoT) technologies. The system ensures extensive and reliable data transfer across rural areas by implementing Long Range (LoRa) technology to establish a Scalable Wireless Sensor Network. Coupled with an integrated alarm system, it markedly improves response times by providing precise, real-time predictions and detections of floods and landslides. The flood prediction model continuously improves through real-time data from IoT devices, enhancing its predictive accuracy. Meanwhile, landslide detection employs image processing techniques to analyze satellite images for early signs such as cracks, and the landslide prediction model provides early warnings. These systems are trained using specific datasets from Sri Lanka, ensuring high relevance and accuracy. Additionally, a web application delivers location-specific disaster notifications, significantly advancing disaster management and preparedness. This approach not only safeguards communities but also protects economic assets, a critical need demonstrated by the devastating 2016 landslides in Aranyaka. [1]**

Keywords—LoRa, IoT, Machine Learning, Community Safety Systems, Remote Sensing, Real-Time Data Processing, Advanced Warning Systems

I. INTRODUCTION

In Sri Lanka, the frequent and severe occurrences of natural disasters like floods and landslides critically endanger human life and economic stability due to the country's specific geographical and climatic conditions. Traditional tools for disaster management often fall short, plagued by inaccurate sensor data and unreliable data transfers, especially in areas with poor cellular coverage.

To address these challenges, this project harnesses the power of the Internet of Things (IoT), machine learning (ML), and image processing technologies. While IoT and Long Range (LoRa) technology are utilized to develop a reliable, scalable wireless sensor network for flood detection, ML is employed to enhance the prediction accuracy of both flood and landslide incidents. Image processing techniques are specifically applied to detect early signs of landslides, such as surface anomalies and shifts detected in satellite images, offering a crucial advantage in early warning systems.

By integrating these technologies, the initiative aims to provide precise, real-time environmental monitoring. This integration enables the system to deliver timely and accurate alerts, significantly improving disaster response and management. The project not only promises enhanced safety for residents in vulnerable areas but also ensures that the system can be adapted and scaled to meet similar challenges in other regions, thereby contributing to broader disaster resilience efforts.

II. LITERATURE REVIEW

The goal of this research "Quantitative Analysis and GIS Integration for Landslide Detection " is to predict the occurrence of landslides by combining quantitative analysis with Geographic Information Systems (GIS). [2] A study which was carried out in Sri Lanka, uses survey data from different universities to pinpoint the main causes of landslides. This research is novel in that it uses GIS data in conjunction with Artificial Neural Networks (ANNs) to improve the dynamic modeling of landslide susceptibility. [3] The approach works well for processing both qualitative and quantitative data, providing a thorough predictive model that accurately and consistently identifies areas at high risk of landslides. This approach underscores the potential of integrating advanced data analytics and spatial information systems in environmental hazard prediction, setting a precedent for future research in the field.

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This study "Transfer Learning and Mask R-CNN in Landslide Segmentation" explores the application of Mask R-CNN and transfer learning to differentiate between recent and past landslide incidents. Using newly collected landslide data, this method applies pre-trained models to identify older, more subtle landslide features. This approach lessens the dependency on sizable, labeled datasets, which are frequently hard to come by in disaster management settings. This technique makes it easier to detect landslides in a variety of geological structures, which speeds up response times and improves preparedness for landslide incidents. The study highlights the flexibility of convolutional neural networks and transfer learning in environmental monitoring, highlighting their potential to enhance landslide detection and response tactics.

The research paper "Modeling of Flood Water Level Prediction Using Improved RBFNN Structure" [4] discusses the use of hydrological sensors to detect floods by continuously monitoring water levels and rainfall. The development of systems that collect vital hydrological data and incorporate it into prediction models to provide timely flood warnings is greatly aided by this research. Effective early warning mechanisms are made possible by the proposed system's ability to predict imminent flood events by analyzing rainfall and water levels using complicated calculations. This study is an example of how advanced predictive analytics, in conjunction with real-time data collection, can greatly improve flood management efforts, potentially saving lives and reducing economic damage in flood-prone areas.

To enhance flood warnings and safety procedures, the research paper "River Water Level and Speed Monitoring and Alert System" attempts to create a real-time monitoring system for evaluating river water levels and flow velocity. This system incorporates an Arduino microcontroller-connected network of ultrasonic and flow rate sensors. The collected data is then transmitted to a central server for processing before being made available on a mobile and web platform to ensure timely notifications and improve response times in the event of a flood, but reliable mobile network coverage is a problem for this system, especially in rural areas. To address this problem, we are implementing a system using LoRa technology, which provides long-range, low-power communication solutions. By extending the operational, LoRa technology can improve connectivity without relying on conventional mobile networks.

III. METHODOLOGY

A. Flood Detection

This study emphasizes the integration of data collection, processing, and communication using Long Range (LoRa) technology in the development of a comprehensive methodology for building a modern flood detection system that utilizes Internet of Things (IoT) technology. Specifically designed for areas with limited access to traditional mobile network coverage, the system offers timely flood alerts and real-time monitoring.

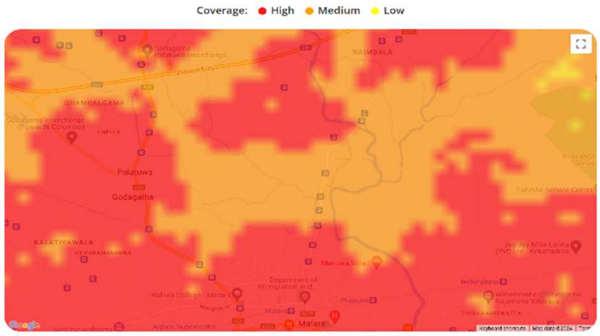

Figure 1: Mobile coverage map near Nilwala river in Srilanka [12]

1) Introduction to LoRa Technology

LoRa (Long Range) is a patented digital wireless data communication IoT technology developed by Cycleo of Grenoble, France, and later acquired by Semtech in 2012. *[5]*

2) Advantages of LoRa Technology

LoRa technology offers extensive range capabilities of up to 10-20 km in rural areas and 2-5 km in urban settings, alongside low power consumption that allows devices to operate for a long time on a single battery charge, making it ideal for IoT devices in remote locations. [6]

System Architecture Overview of IoT device

1) Data Collection and Sensor Implementation: The system, featuring high-precision water level sensors are deployed along the Nilwala River—known for frequent and sudden flood events—based on historical data and in collaboration with the National Building Research Organization (NBRO), with sensors installed at strategic points.

2) Sensor Details: The WagonR Ultra Sonic water level sensor, known for its waterproof capabilities and ability to detect minimal variations in water levels, is used.

3) Setting Up a LoRa Communication Network: Each sensor is equipped with a LILYGO SX1268 LoRa module [Figure 2.1] , transmitting data to a central LoRa base station powered by an SX1302 module connected to a Raspberry Pi 4, which serves as the communication hub using SF12.

Figure 2.1: LILYGO SX1268

Figure 2.2: SX1302 module

4) Energy Efficiency: Considering the remote sensor locations, the system is designed with energy-efficient components to minimize power consumption, ensuring long operational periods without the need for frequent maintenance.

5) Operational Results:

A range test was conducted in Sri Lanka Institute of Information Technology as shown in the Figure 3, the Lora gateway receiver was placed on the sixth floor of main building with an altitude of 21m, approximately. The gateway is powered and connected to the internet.

The end-node device was relocated along various location points in the surrounding area. It periodically sent data on the dummy water level data and real-time GPS coordinates. A screenshot of the situational map is shown in Figure 3, with the location of the LoRa gateway and several measurement points from P1 to P2 highlighted. From each of these positions, the end-node and the gateway exchanged many messages.

Figure 3: P1 to P2 represent the measurement locations. The gateway location is represented by a green icon and the end node is represented by a red icon.

Table 1: Distance, SNR for each outdoor location

Location	Distance [m]	SNR	RRSI
P1 [6°54'53.3"N 79°58'21.9"E]	71.9328	8.5	-83
P2 [6°54'54.3"N 79°58'28.0"E]	321.869	6.5	-91

The LoRa signal's ability to adapt to different environmental conditions and distances can be observed through the range test conducted between two locations, P1 and P2. The signal quality (SNR) and strength (RSSI) at each location are summarized in Table 1, along with the corresponding distances from the base station. At a close range of 71.93 meters, P1, with a reasonably clear line of sight but some tree interference, maintains a strong RSSI of -83 dBm and a high SNR of 8.5. At 321.86 meters, P2, on the other hand, exhibits a weaker signal with an RSSI of -91 dBm and a lower SNR of 6.5 due to buildings and other obstructions.

Table 2:Minimum SNR requirements of LoRa SFs for successful communication. [7]

SF	SF7	SF8	SF9	SF10	SF11	SF12
Required SNR	-6.5 dB	-8.5 dB	-11 dB	-13.5 dB	-18.5 dB	-21 dB

Referencing the SNR requirements from Table 2, our results at location P2 with a SNR of 6.5 dB greatly exceed the minimum required SNR of -21 dB for SF12, suggesting significant potential for extending operational distances. This indicates that our LoRa system could effectively function at distances far beyond the 321.87 meters tested, particularly in similar environmental conditions.

B. Landslide Detection

In this study, we developed a landslide detection system for the Aranyaka region in Sri Lanka using remote sensing data and deep learning techniques. The data utilized in this study included multispectral images from Sentinel-2, covering bands B1 to B12, slope data from ALOS PALSAR (B13), and digital elevation model (DEM) data from ALOS PALSAR (B14). All bands were resampled to a resolution of approximately 10 meters per pixel, and image patches were standardized to 128x128 pixels with pixel-wise labeling to prepare the data for model training.

Preprocessing involved resizing and normalizing the satellite images to ensure consistent input dimensions and to enhance the model's learning capabilities. Data augmentation techniques, including rotation, flipping, and zooming, were employed to increase the diversity of the training data and improve the model's ability to generalize. Binary masks indicating landslide areas were generated and aligned with the satellite images to serve as labels for the training process.

U-Net was first applied to biomedical image segmentation, followed by numerous semantic segmentation applications that demonstrated successful results. [8] This model has since been widely adopted for various tasks, including landslide detection. [9] The U-Net architecture was selected for this study due to its effectiveness in semantic segmentation, which is crucial for identifying landslide-prone areas. U-Net's encoder-decoder structure is particularly adept at capturing both low-level and high-level features through its use of skip connections.

The encoder path consists of a series of convolutional layers with a kernel size of 3x3, each followed by a rectified linear unit (ReLU) activation function. After every two convolutional layers, max-pooling layers with a filter size of 2x2 and a stride of 2 are employed to progressively down sample the feature maps. Additionally, dropout layers are integrated within the convolutional blocks to mitigate overfitting and enhance model robustness.

The decoder path mirrors the encoder but employs transposed convolutional layers for up sampling. These layers increase the spatial resolution of the feature maps while restoring the fine-grained details lost during the down sampling process. Skip connections link corresponding layers in the encoder and decoder, allowing the model to merge detailed spatial information from the encoder with the high-level features in the decoder. The final layer of the model uses a 1x1 convolution with a sigmoid activation function to generate a binary mask, predicting the likelihood of landslide occurrence at each pixel.

The U-Net model implemented in this study comprises a total of 19 convolutional layers and 3 transposed convolutional layers. The model was compiled with the Adam optimizer and trained using binary cross-entropy loss, alongside custom metrics including F1 score, precision, and recall assessing its performance.

For training, the dataset was split into 80% for training and 20% for validation. Key hyperparameters included a batch size of 16, 150 epochs, and the Adam optimizer. The model was trained using TensorFlow on Google Colab, leveraging a T4 GPU to accelerate the process. A custom Dice loss function was incorporated to address class imbalance and to improve the accuracy of segmentation [10], particularly in the detection of smaller landslide areas.

C. *Landslide and Flood Prediction*

1) *Data Collection and Preparation*

1. Flood Prediction Data: Data was gathered from an extensive meteorological database that included monthly rainfall data for each of Sri Lanka's 26 districts from 2014 to 2024. Additionally, daily water levels from 2018 to 2024 were acquired from the Disaster Management Center. This dataset was randomly split into 70% for training and 30% for testing the model.

2. Landslide Prediction Data: Data was collected from reports generated by the National Building Research Organization (NBRO), focusing on sites within Giddawa, Victoria, Warakapola, Madulkelle, Matale-P.W.D, Ibbagamuwa, Kotmale Power STN-D/S, Maskeliya Hospital areas in Sri Lanka's Central Province. The dataset was similarly divided into 70% for training and 30% for testing. The landslide conditioning factors considered were topographic (slope angle, elevation), geological [8] (soil type, geology), hydrological (groundwater, hydrology), climatic (rainfall), land use (land use and vegetation), spatial (longitude, latitude), and subsurface (overburden thickness).

Table 3:Landslide Conditioning Factors and Data Types

Geographical feature	Conditioning Factor	Type of Data
Topographic Factors	Slope Angle	Numeric
	Elevation	Numeric
Geological Factors	Soil Type	Categorical
	Geology	Categorical
Hydrological Factors	Hydrology	Numeric
Climatic Factors	Rainfall	Numeric
Spatial Factors	Longitude	Numeric
	Latitude	Numeric
Subsurface Factors	Overburden Thickness	Numeric

2) *B. Feature Selection and Data Analysis*

1. Flood Prediction Features: For flood prediction features, data preprocessing involved scaling numerical features such as rainfall amounts and water levels and encoding categorical features. Outlier detection was performed using box plots to identify and handle outliers, ensuring robust and accurate predictions. The model training and evaluation phase utilized a Random Forest Classifier, which was assessed using metrics such as accuracy, precision, recall, F1 score, and ROC AUC score. Additionally, precision-recall analysis was conducted to determine the optimal threshold for predicting floods

2. Landslide Prediction Features: For landslide prediction features, correlation analysis was conducted using a correlation matrix heatmap to identify relationships between numeric conditioning factors, highlighting highly correlated factors. Feature selection employed Information Gain and Information Gain Ratio to quantify the predictive ability of causative factors, removing those with null predictiveness. The model training and evaluation phase implemented a Random Forest algorithm for landslide susceptibility prediction, with performance evaluated using sensitivity, specificity, accuracy, kappa index, and the area under the ROC curve.

3) *Model Implementation*

1. Flood Prediction Model: The system combines Random Forest (RF) for classifying water levels into flood categories and Long Short-Term Memory (LSTM) for predicting future flood risks. RF employs a forest of decision trees trained on varied data subsets for robust classification [3]. Meanwhile, LSTM captures temporal dependencies in sequential data like water levels and rainfall, enabling accurate flood forecasting tailored to real-time scenarios.

The model integrates cost-effective sensors to monitor environmental parameters such as water levels and rainfall in flood-prone areas. These sensors ensure compatibility and optimal performance with the machine learning model tailored to them [2]. The data collected in real-time from IoT devices is processed by the Random Forest Classifier for classification and by an LSTM model for predicting future flood risks.

2. Landslide Prediction Model: This model is built using a comprehensive machine learning pipeline, combining data preprocessing, feature engineering, and model training to deliver accurate predictions. Historical and real-time data on rainfall, slope angle, elevation, and overburden thickness were used as key predictors. A custom function enabled dynamic data retrieval from the Open-Meteo API, ensuring the model adapts effectively to changing environmental conditions. The preprocessing stage involved encoding categorical features like overburden thickness and hydrology using custom encoders and scaling numerical variables, such as latitude, longitude, elevation, and average temperature, for consistency and improved learning outcomes.

The pipeline incorporated multiple machine learning models: Linear Regression, Random Forest (RF), and Gradient Boosting. Linear Regression served as a baseline, establishing fundamental relationships among key predictors and trends. RF utilized ensemble learning to improve prediction accuracy, reducing overfitting and capturing complex nonlinear relationships in the data. Gradient Boosting provided an advanced, iterative approach, sequentially refining predictions to minimize errors and achieve superior performance in capturing intricate data patterns.

The models were trained on features such as one-hot-encoded area identifiers, hydrology patterns, and scaled numerical factors. Through feature importance analysis, rainfall, elevation, latitude, and longitude emerged as significant predictors of landslide susceptibility. Employing these models enabled a detailed comparison of their performance, ensuring the final solution is robust and reliable.

4) D. Performance Evaluation

The performance of both prediction models was evaluated using comprehensive metrics:

- Flood Prediction: Accuracy, precision, recall, F1 score, and ROC AUC score.
- Landslide Prediction: Sensitivity, specificity, accuracy, kappa index, and the area under the ROC curve.

By combining these methodologies, our research aims to develop robust, accurate, and real-time predictive models for both floods and landslides, tailored to the specific conditions and requirements of Sri Lanka.

IV. Results and Discussion

The U-Net model's performance was evaluated across several key metrics, including loss, accuracy, precision, recall, and F1 score, with a threshold of 0.5 applied to the model's output to classify landslide-prone areas. After 150 epochs, the model achieved a training loss of 0.0435, an accuracy of 0.9876, an F1 score of 0.7033, a precision of 0.8032, and a recall of 0.6313. These results indicate strong overall performance, with the model effectively identifying landslide-prone areas, although the slightly lower recall suggests that some landslide areas might have been missed. The performance metrics for both the training and validation datasets are visually represented in Figure 4, which shows the model's loss, precision, recall, and F1 score over the course of the training process.

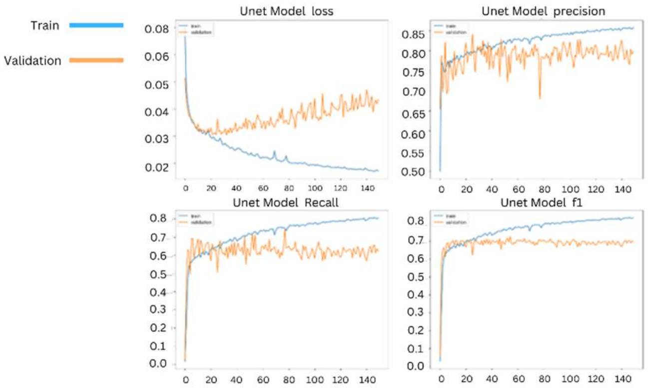

Figure 4: Performance metrics during training and validation (loss, precision, recall, and F1 score).

To better understand the model's predictions, a threshold of 0.5 was used to convert the continuous output probabilities into binary classifications, where predictions greater than 0.5 were considered positive (indicating a landslide) and those less than or equal to 0.5 were considered negative. Visual inspection of the model's predictions provides further insights into its effectiveness. Two representative examples are shown in Figures 5 and 6. Figure 5 demonstrates a successful detection scenario where the model accurately identifies landslide-prone areas. The predicted landslide regions, when compared to the ground truth labels, show a strong correspondence, as depicted in the respective images. Conversely, Figure 6 presents a case where the model correctly identifies the absence of a landslide, with both the prediction and the ground truth indicating no detected landslide areas.

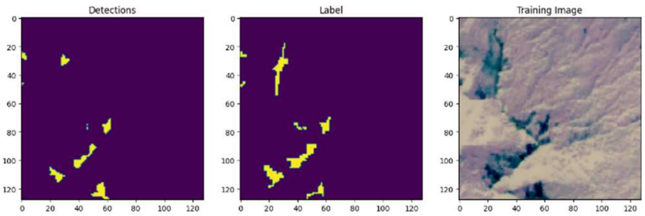

Figure 5: Successful landslide detection with the model's predictions, ground truth labels, and the corresponding satellite image.

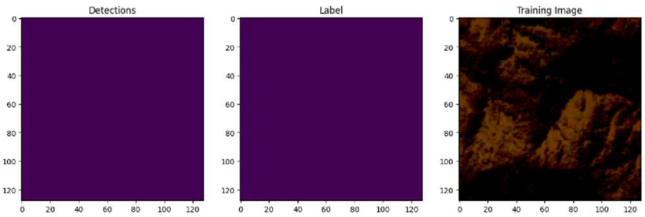

Figure 6: Example of no landslide detection, with the model's predictions, ground truth labels, and the corresponding satellite image.

These results collectively highlight the U-Net model's capability in landslide detection, with a good balance between precision and recall. However, there remains room for improvement, particularly in enhancing recall to ensure that fewer landslide areas are missed. Future enhancements, such as incorporating attention mechanisms or experimenting with alternative architectures, could further improve the model's performance in complex and varied environmental conditions.

The results show that the LoRa signal strength and quality vary significantly between the two tested locations. At location P1, despite minor obstructions, the signal remained robust with an RSSI of -83 dBm and an SNR of 8.5. Conversely, at location P2, which faced more significant barriers such as buildings and was further from the gateway, the signal weakened, with an RSSI of -91 dBm and an SNR of 6.5. These observations underline the influence of environmental conditions and distance on LoRa communication effectiveness.

The landslide prediction model demonstrated strong performance across key metrics, underscoring its utility for disaster management in Sri Lanka. Rainfall was identified as the most critical factor influencing landslide susceptibility, followed by average temperature, elevation, and hydrological patterns. Spatial features, such as latitude and longitude, also contributed but to a lesser extent.

The model achieved an overall accuracy of 84%, with an F1 score of 0.78, indicating a good balance between precision and recall. Receiver Operating Characteristic (ROC) curves further highlighted the model's predictive capability, with AUC values ranging from 0.85 to 0.89. These results confirm the model's reliability in distinguishing between landslide-prone and non-prone areas.

Real-time predictions were validated using environmental and rainfall data from regions such as Ibbagamuwa and Madulkelle. The model estimated 2–5 potential landslides in these areas, aligning with historical trends. By dynamically incorporating real-time data, the model provided actionable insights, enhancing its practical application for early warning systems.

The flood prediction model, utilizing a Random Forest Classifier, achieved an accuracy of 80% on the test set. It performed strongly in predicting "normal" and "minor flood" conditions, with F1 scores of 0.85 and 0.89, respectively. The model showed promising results for the "alert" category, with an F1 score of 0.75. The overall weighted average F1 score across all categories was 0.86, indicating robust general effectiveness. The LSTM model achieved 85% accuracy.

Table 4:Classification Report

	Precision	Recall	F1 Score	Support
Normal	1.0	1.0	1.0	24.0
Flood	8.33333	1.0	0.909	5.0
Alert	0.66666	0.4	0.5	5.0
Accuracy	0.88095	0.8809	0.88	0.88
Macro Avg	0.63333	0.6514	0.63	42.0
Weighted Avg	0.86111	0.8809	0.864	42.0

Cross-validation results, depicted in Figure 9, illustrate consistent improvements in performance metrics—accuracy, precision, recall, and F1 score—across the folds, achieving up to 0.80 accuracy in the final fold. These results suggest that the model generalizes well and may benefit from further refinement to enhance predictions for the "alert" category.

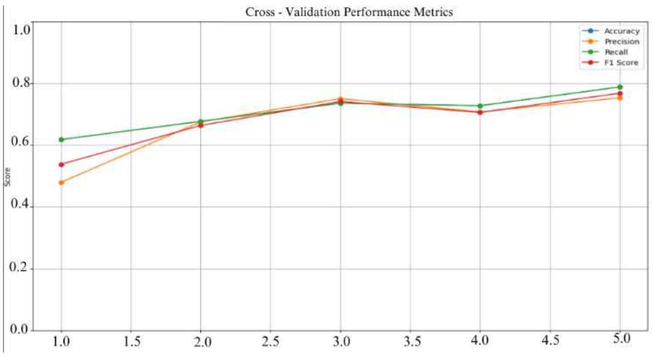

Figure 9: Cross Validation Performance Metrics

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